SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B.Tech- AIDS/2024	**Date:**	03/08/2026

Code of Conduct

I S.Krishna Chaithanya(192424287)*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **S.Krishna Chaithanya**

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.

1. Morphological Analysis Pipeline for Document Indexing System

Problem Understanding

The words connected, connecting, connection belong to the same word family. A morphological analysis pipeline separates each word into its root (stem) and suffix, identifies whether the suffix is inflectional or derivational, and normalizes all forms to a common base for efficient document retrieval.

Method Selection

Use a rule-based morphological analyzer with the following steps:

1. Tokenization – Read input words.
2. Suffix Detection – Identify known suffixes (-ed, -ing, -ion).
3. Root Extraction – Remove the suffix and recover the root.
4. Morphological Classification – Determine inflectional or derivational form.
5. Normalization (Lemmatization) – Convert all variants to the common base form connect.
6. Indexing – Store the normalized form for searching.

Solution Process

Input Word	Root	Suffix	Morphological Type	Parsed Structure	Normalized Output
connected	connect	-ed	Inflectional (Past tense)	connect + ed	connect
connecting	connect	-ing	Inflectional (Present participle)	connect + ing	connect
connection	connect	-ion	Derivational (Forms noun)	connect + ion	connect

Accuracy of Calculation

- connected → connect + ed ✓ Correct (Inflectional)
- connecting → connect + ing ✓ Correct (Inflectional)
- connection → connect + ion ✓ Correct (Derivational)
- All words are successfully normalized to the base form connect, ensuring accurate indexing and retrieval.

Interpretation of Results

- Inflectional forms (connected, connecting) change only the grammatical form of the verb without changing its core meaning.
- Derivational form (connection) changes the word class from a verb to a noun.
- After normalization, all variants are indexed as connect, allowing a search for connect to retrieve documents containing connected, connecting, and connection, thereby improving search efficiency and recall.

2. Morphological Parsing Module for Sentiment Analysis

Problem Understanding

The words unhappy, happiness, happily belong to the same word family and share the common base word happy. A morphological parsing module identifies the prefix, root (base form), and suffix of each word, determines whether the transformation is inflectional or derivational, and ensures that all related forms are mapped to a common root for consistent sentiment analysis**.**

Method Selection

Use a rule-based morphological parser with the following steps:

- Tokenization – Read the input words.
- Prefix Detection – Identify prefixes such as un-.
- Suffix Detection – Identify suffixes such as -ness and -ly.
- Root Extraction – Remove prefixes and suffixes to obtain the base word.
- Morphological Classification – Determine whether the transformation is inflectional or derivational.
- Normalization – Convert all variants to the common base form happy.
- Sentiment Mapping – Use the normalized root for consistent sentiment analysis.

Solution Process

Input Word	Prefix	Root	Suffix	Morphological Type	Parsed Structure	Normalized Output
unhappy	un-	happy	—	Derivational (Negative adjective)	un + happy	happy
happiness	—	happy	-ness	Derivational (Forms noun)	happy + ness	happy
happily	—	happy	-ly	Derivational (Forms adverb)	happy + ly	happy

Accuracy of Calculation

- unhappy → un + happy ✓ Correct (Derivational)
- happiness → happy + ness ✓ Correct (Derivational)
- happily → happy + ly ✓ Correct (Derivational)
- All words are successfully normalized to the base form happy, ensuring semantic consistency for sentiment analysis.

Interpretation of Results

- Unhappy is formed by adding the prefix un-, which changes the meaning of happy to its opposite, making it a derivational form.
- Happiness is formed by adding the suffix -ness, changing the word class from an adjective to a noun, making it a derivational form.
- Happily is formed by adding the suffix -ly, changing the word class from an adjective to an adverb, making it a derivational form.
- After normalization, all variants are mapped to the root happy, enabling the sentiment analysis system to recognize related words consistently and improve sentiment classification accuracy.

3. Stemming-Based Preprocessing Module for a Search Engine

Problem Understanding

The search engine receives different forms of the same word:

- played
- player
- playing

These words should be reduced to a common base so that searching for one form retrieves documents containing the others. A stemming module removes common prefixes/suffixes to produce a consistent root.

Method Selection

Technique: Stemming (Suffix Stripping)

Morphological Rules:

1. Remove -ed → indicates past tense.
2. Remove -ing → indicates present participle/continuous form.
3. Remove -er → indicates agent/person performing an action.
4. Extract the common stem.

Solution Process

Input Word	Morphological Rule Applied	Extracted Stem	Classification
played	Remove -ed	play	Past Tense Verb
player	Remove -er	play	Agent Noun
playing	Remove -ing	play	Present Participle Verb

Algorithm

1. Read the input word.
2. Check for suffixes:
    - If word ends with "ing", remove ing.
    - Else if word ends with "ed", remove ed.
    - Else if word ends with "er", remove er.
3. Store the remaining part as the stem.
4. Classify the word based on the removed suffix.
5. Return the stem and classification.

Accuracy of Calculation

- played → play ✓ Correct
- player → play ✓ Correct
- playing → play ✓ Correct

All three words are successfully reduced to the same stem "play", ensuring consistent indexing in the search engine.

Interpretation of Results

The stemming module extracts the common root "play" from all input words. Although the words have different grammatical meanings (past tense, agent noun, and present participle), they are indexed under the same stem. This improves search efficiency by matching all related word forms during retrieval.

Final Result

Input Word	Stem	Classification
played	play	Past Tense Verb
player	play	Agent Noun
playing	play	Present Participle Verb

4.Finite-State Morphological Parsing Module for Text Processing System

Problem Understanding

The goal is to design a finite-state morphological parsing module for the words writes, writing, and written. The module uses state transitions to identify suffixes and irregular verb forms, classifies them as regular or irregular inflectional patterns, and maps all words to the common root write.

Method Selection

Use a Finite-State Morphological Parser (FSM) with the following steps:

1. Read the input words.
2. Apply finite-state transitions to detect suffixes and irregular forms.
3. Decompose each word into root + suffix.
4. Classify each transformation as regular or irregular inflectional.
5. Normalize all words to the common root write.

Solution Process

- writes → Detect suffix -s, remove it, extract root write.
- writing → Detect suffix -ing, remove it, extract root write.
- written → Recognize the irregular form -en, map it to the root write using FSM rules.
- Normalize all forms to the base word write.

Accuracy of Calculation

Input Word	State Transition	Morphological Breakdown	Classification	Normalized Output
writes	write → writes	write + s	Regular Inflection	write
writing	write → writing	write + ing	Regular Inflection	write
written	write → written	write + en (Irregular)	Irregular Inflection	write

Interpretation of Results

The finite-state morphological parser successfully extracts the common root write from all input words. Writes and writing follow regular inflectional patterns, whereas written is an irregular inflectional form. By normalizing all variants to write, the system ensures consistent indexing, text processing, and information retrieval.

5. Porter Stemmer-Based Preprocessing Module for Document Retrieval

Problem Understanding

The goal is to design a Porter Stemmer-based preprocessing module for the words relational, relation, and relate. The module applies Porter stemming rules to remove derivational suffixes, preserve semantic consistency, and normalize all words to a common stem for efficient document retrieval.

Method Selection

Use the Porter Stemming Algorithm with the following steps:

1. Read the input words.
2. Apply Porter stemming rules to remove derivational suffixes.
3. Generate intermediate forms after each stemming step.
4. Extract the common stem.
5. Normalize all words to the final stem.

Solution Process

- relational → Remove -ational → relate → Remove -e → relat
- relation → Remove -ion → relat
- relate → Remove -e → relat
- Normalize all words to the common stem relate

Accuracy of Calculation

Input Word	Stemming Steps	Intermediate Form	Final Stem
relational	Remove -ational, then -e	relate	relat
relation	Remove -ion	relat	relat
relate	Remove -e	relat	relat

Interpretation of Results

The Porter Stemmer successfully reduces relational, relation, and relate to the common stem relat. Although relat is not a complete English word, it effectively groups all related terms under a single index, improving document retrieval efficiency and search performance.

Rubrics for Calculation

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	**Problem Understanding**	**Method Selection**	**Solution Process**	**Accuracy of Calculation**	**Interpretation of Results**	**Sub. Total**	**Total %**
1							
2							
3							
4							
5							

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________